

CAPSTONE PROJECT EVALUATION: REVIEW 1 REPORT (TOTAL: 20 MARKS)

GenVendorAI: AI-Powered Vendor Intelligence & Risk Assessment System

Comprehensive Problem Identification, Need Analysis, Objectives, Methodology & Feasibility Report

Evaluation Criterion (Review 1 Rubrics)	Max Marks	Deliverables / Section Mapping
1. Problem Identification & Relevance	5 Marks	Section 1: Enterprise MDM Challenges, Fraud Risks & Statutory Background
2. Literature Review & Need Analysis	5 Marks	Section 2: Comparative Literature Survey & 4 Core Research Gaps
3. Objectives, Methodology, Feasibility & Planning	5 Marks	Section 3: SMART Objectives, Theoretical Methodology & Gantt Schedule
4. Synopsis, Documentation & Presentation	5 Marks	Section 4: Preliminary Architecture, Mathematical Model & Documentation
TOTAL EVALUATION WEIGHTAGE	20 MARKS	Full Capstone Phase 1 Milestone Verification

Project Group ID: 104	Project Guide: Dr. Ashwini Deshpande
Team Members: Chirag Jathe (123B1E022), Prashik Dekate (123B1E025), Sharad Gajjewar (123B1E038), Bhakti Kulkarni (123B1E058)	Target Tech Stack: FastAPI, Sentence-BERT, FAISS, PyMuPDF, Tesseract OCR, SQLite

RUBRIC 1: PROBLEM IDENTIFICATION & RELEVANCE [5 MARKS]

1. Problem Context, Enterprise Relevance & Motivation

1.1 Industrial Context: In contemporary mid-sized and large-scale enterprises, Vendor Master Data Management (MDM) serves as the backbone of financial operations, supply chain logistics, and statutory tax reconciliation. Organizations utilizing ERP suites (SAP S/4HANA, Oracle ERP Cloud, Microsoft Dynamics) must register, verify, and monitor thousands of external contractors, logistics carriers, and material suppliers.

1.2 Problem Definition: Currently, the onboarding lifecycle is governed by fragmented manual workflows where vendors submit unstandardized PDF documents (GST certificates, PAN cards, bank verification letters, and quotations). This manual reliance causes severe enterprise liabilities:

- Turnaround Latency:* Manual validation of statutory credentials consumes 45 to 90 minutes per vendor, causing 7-to-14-day onboarding backlogs.
- Data Transcription Errors:* Human entry of 15-character GSTINs, 10-character PANs, and 11-character IFSC codes suffers an 8%–12% error rate, triggering rejected invoice payments and delayed tax credit claims.
- Shell Entity & Sister-Concern Fraud:* Fraudulent suppliers register multiple shell entities sharing identical addresses or bank accounts to circumvent single-supplier bidding ceilings.
- Statutory Compliance Vulnerability:* Transacting with non-compliant or mismatched GSTIN profiles exposes enterprises to severe financial penalties under Indian tax statutes.

1.3 Relevance to AI & ML: This problem presents a compelling multidisciplinary challenge combining multi-modal computer vision (OCR document decoding), natural language layout parsing, dense vector similarity representation (Sentence-BERT), approximate nearest neighbor indexing (FAISS), and explainable risk decision theory.

RUBRIC 2: LITERATURE REVIEW & NEED ANALYSIS [5 MARKS]

2. Literature Survey, Comparative Study & Research Gaps

An extensive literature survey was conducted across document processing, record linkage, and automated risk scoring models:

Approach / Prior Art	Underlying Methodology	Identified Deficiencies & Research Gaps
Rule-Based Regex & Template Matching	Hardcoded bounding boxes and regex patterns looking for colon-separated labels.	Extremely brittle; completely breaks on borderless tables, tab-delimited columns, or scanned photocopies lacking colons.
Fuzzy String Distance (Levenshtein / Jaro-Winkler)	Lexical character edit distance measurement between entity strings.	Semantically blind; cannot recognize word reordering, abbreviations (e.g. 'Pvt Ltd' vs 'Private Limited'), or conceptual sister companies.

Cloud LLM APIs (OpenAI / Gemini)	Sending document text prompts to remote commercial foundation models.	Prohibitive per-token recurring costs, high latency (3-10s), and critical enterprise data leakage violating India's DPDP Act 2023.
Black-Box Neural Classifiers	Deep neural networks predicting binary approved/rejected classifications.	Lacks statutory explainability; fails corporate audit requirements by not providing itemized positive and negative compliance evidence.

Need Analysis: Enterprises urgently require an integrated, local-first intelligence architecture that unifies OCR, table parsing, statutory checksum checks, dense semantic vector search, and transparent explainable decision support into a sub-second autonomous pipeline.

RUBRIC 3: OBJECTIVES, METHODOLOGY, FEASIBILITY & PLANNING [5 MARKS]

3. Project Objectives, Proposed Methodology & Project Schedule

3.1 SMART Objectives:

- 1. Achieve >=98% field-level extraction accuracy across 11 canonical vendor master attributes.
- 2. Implement 100% deterministic statutory Indian validation (GSTIN state codes, PAN legal entity decoding, RBI IFSC).
- 3. Detect exact database duplicates and semantic near-duplicates with >=90% precision using SBERT + FAISS.
- 4. Formulate an explainable AI risk scoring model (0–100 scale) outputting auditable positive and risk findings.
- 5. Deliver a production-grade full-stack web application running 100% offline with sub-second execution latency.

3.2 Feasibility Analysis:

- *Technical Feasibility:* Verified using Sentence-BERT (MiniLM-L6-v2) and FAISS running locally on standard CPU hardware (16GB RAM) without requiring GPUs.
- *Economic Feasibility:* 100% open-source stack (FastAPI, SQLite, PyMuPDF, Tesseract, ReportLab) guarantees zero recurring cloud API subscription costs.
- *Operational Feasibility:* Intuitive modern SPA frontend usable by procurement personnel without technical machine learning expertise.

Project Phase / Milestone	Planned Timeline	Target Deliverables
Phase 1: Synopsis & Requirement Analysis	Weeks 1 – 4	Literature survey, problem definition, architecture specification
Phase 2: Ingestion & Layout Extraction Engine	Weeks 5 – 8	PyMuPDF digital stream parser, Tesseract LSTM OCR, Table-Aware KV parser
Phase 3: Statutory Validator & Vector Store	Weeks 9 – 12	GSTIN/PAN/IFSC validator, SBERT 384-D encoder, FAISS IndexFlatIP store
Phase 4: Risk Scoring & Decision Support	Weeks 13 – 16	Explainable 0-100 penalty scoring, ReportLab PDF dossier compiler
Phase 5: Full-Stack Web App & Deployment	Weeks 17 – 20	FastAPI REST backend, Tailwind CSS SPA, comprehensive testing suite

RUBRIC 4: SYNOPSIS, DOCUMENTATION & PRESENTATION [5 MARKS]

4. Preliminary Architectural Design & Mathematical Formulation

4.1 Mathematical Foundations:

- *Dense Representation:* Vendor profile text mapped to $\mathbf{v}$ in $\mathbb{R}^{384}$ via SBERT.
- *Cosine Similarity in FAISS:* $\text{Cosine_Sim}(\mathbf{u}, \mathbf{v}) = (\mathbf{u} \cdot \mathbf{v}) / (\|\mathbf{u}\|_2 \cdot \|\mathbf{v}\|_2) = \sum_{i=1}^{384} (u_i \cdot v_i)$
- *Explainable Risk Formulation:* $\text{Risk_Score} = \text{Min}(100, \sum w_i \cdot I(\text{Exception}_i) + w_{\text{dup}} \cdot I(\text{Duplicate}) + w_{\text{sim}} \cdot \text{Similarity_Penalty})$

4.2 Documentation Deliverables: Complete project synopsis, architectural data-flow diagrams, relational schemas, and initial prototype validation scripts compiled and verified in D:\GenVendorAI.